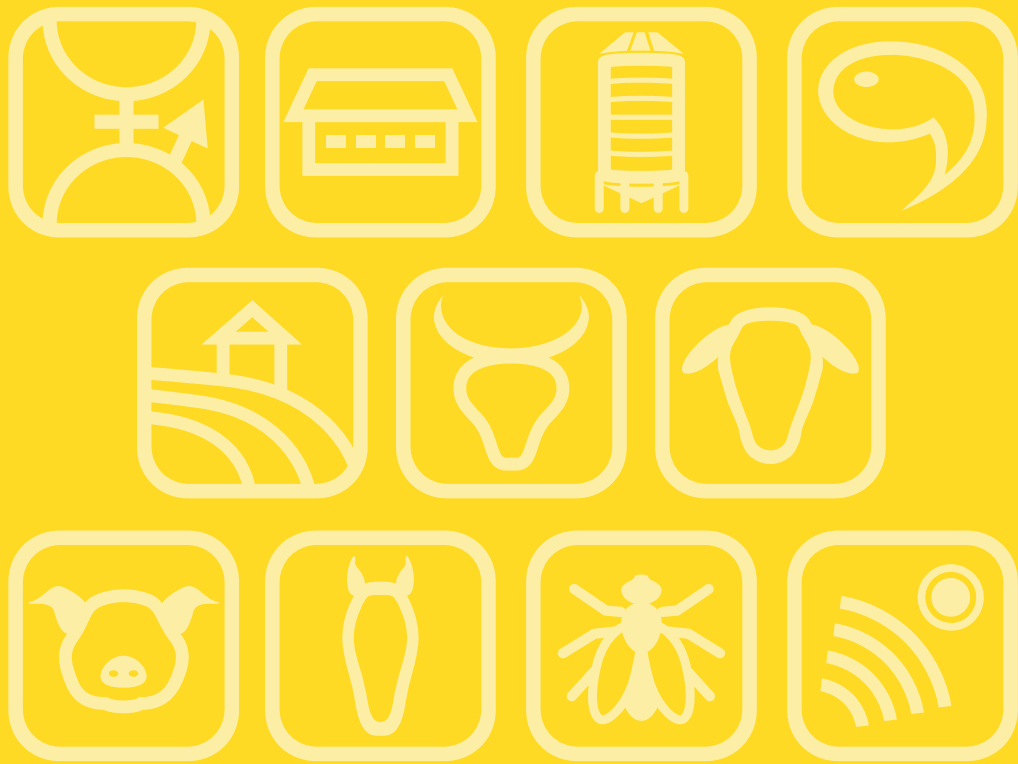


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Genetic parameters of fibre density in alpacas

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Fleece weight determines the income of the producer. Fibre diameter is the most commonly used selection criterion in alpacas, and this leads to an undesirable response in fleece weight due to the positive genetic correlation between them. However, higher fibre density increases the number of fibres compensating their lower weight, leading to heavier fleeces. Fibre density can be addressed by a subjective scoring method from 1 to 5, but also by a density quantification device (Fiber-DEN), where a 1 cm² area is shaved at the midline level and the skin dyed, differentiating very clearly the fibre follicles in white alpacas. This study aims to estimate the genetic parameters of hair duct number (HD), total fibre number (TF) and HD/TF ratio (RT) measured by Fiber-DEN and their relationship with the main textile traits such as fibre diameter (FD), medullation percentage (PM), greasy fleece weight (GFW), subjective density score (DE) and staple length (SL). A total of 402 fibre-DEN records of white Huacaya alpacas aged 1 to 16 years were used along with 8763 (FD and PM), 5752 (DE) and 11271 (GFW and SL) records for other traits. The pedigree consisted of 15 360 animals. The heritabilities were 0.40, 0.47 and 0.37 respectively for HD, TF and RT, and 0.31, 0.28, 0.16, 0.34 and 0.05 for FD, PM, GFW, DE and SL. The relevant genetic correlations between fibre-DEN traits and others were -0.50 for HD-FD and TF-FD, 0.36 and 0.43 respectively for GFW-TF and GFW-RT, and 0.50 for DE-GFW. DE and TF might be used as selection criteria to simultaneously reduce FD and increase GFW in a genetic improvement program, the latter being more costly

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Time budgets in stabled dairy dromedary camels

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Time budgets is used to determine the amount of time camels allocate to different behaviors throughout the day and can be a useful tool to evaluate animals' welfare and management systems. This work investigated the time budgets activities and their implications on the welfare of intensively managed dairy camels. Nine clinically healthy Maghrebi camels (aged 11.7 ± 1.9 years, weighing 437.8 ± 12.0 kg and 72.6 ± 7.1 DIM) housed in a loose stall barn were monitored for 6 consecutive days. Major behavioral activities such as standing, lying, walking, feeding, rumination, milking, as well as other activities like coprophagy, suckling, and sleeping, were continuously recorded. The mean duration of these activities was calculated to obtain the time budget. Results showed that camels spent most of their time in a standing position feeding ($P < 0.0001$). Specifically, camels were standing for 54% of their time and lying down for 38%. Data revealed that the main behavioral activities expressed were feeding (35.5%) followed by rumination (24.5%). Walking activity remained low throughout the day, representing only 2% of the total time budget of stabled camels. No common stereotypic behaviors were recorded; however, coprophagy was detected and recorded 10.3 ± 1.7 min/day. These results suggest that time budgets should be taken into consideration to improve management conditions and welfare status in stabled dairy camels.